Math Inquirers

Extension Workbook

with answers

Level 4

Self-study | Ages 9-11



My name _____

About this book

Math Inquirers Extension Workbook, Level 4
Self-study practice with answers.

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Good to know

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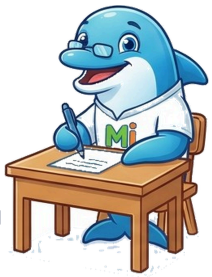
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Factor hunters	Not in this proof edition				☐
2	Big number puzzles	Not in this proof edition				☐
3	Operation detectives	Not in this proof edition				☐
4	Fraction frontier	Not in this proof edition				☐
5	Decimal detectives	Not in this proof edition				☐
6	Angle turns	Not in this proof edition				☐
7	Area architects II	Not in this proof edition				☐
8	Symmetry and nets	Not in this proof edition				☐
9	Data detectives	Not in this proof edition				☐
10	Puzzle Day	6-7	8-9	10 and 12	11	☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

Learn it: Mixed puzzle heuristics

My goal. I can choose a way in to a puzzle: guess and check, suppose all are the same, work backwards, draw it or use a logic grid.

A puzzle needs a way in. Guess and check with a table: use each try to decide whether to go up or down. Suppose all are the same: then see how many swaps you need to fix the total.

Worked example 1

A shop has stools (3 legs) and chairs (4 legs). There are 11 of them with 38 legs in all. How many chairs are there?

Suppose all 11 are stool	
Legs	11 × 3 = 33
Legs short	38 − 33 = 5
1 swap adds	1 leg

1. Each swap from a stool to a chair adds 1 leg.
2. 5 legs short needs 5 swaps.

Answer: 5 chairs and 6 stools. Check: 5 × 4 + 6 × 3 = 38.

Worked example 2

Two next-door numbers multiply to 380. What are they?

Try	Product	Check
18 × 19	342	too small
20 × 21	420	too big
19 × 20	380	right

1. Try a pair. Is the product too big or too small?
2. Use each try to choose the next one.

Answer: The numbers are 19 and 20.

Ways in

1. Guess and check: keep a table of tries.
2. Suppose all are the same. Count swaps.
3. Work backwards: undo the last step first.

Good to know

- Draw a queue or a bar model.
- Check the answer against every clue.

Your turn

Answer at the back of the book.

Try **A box has bees (6 legs) and spiders (8 legs). There are 10 animals with 66 legs.**

How many spiders are there? ☐

Start here

Do these first. Take your time.

1 **Two next-door numbers multiply to make 506. What are they? Use the table for your tries.**

First number	Next number	Product	Too big or small?

The numbers are ☐ and ☐

2 **Mei buys some $3 pens and some $5 notebooks. She buys at least 1 of each. She pays exactly $41.**

(a) How many different ways can she do it? ☐

(b) The way with the fewest pens: ☐ pens and ☐ notebooks

Notebooks, money left, pens: _____

3 **A farm has hens (2 legs) and cows (4 legs). There are 17 animals with 50 legs.**

(a) Suppose all 17 are hens. They have ☐ legs.

(b) That is ☐ legs short.

(c) Each swap to a cow adds 2 legs. There are ☐ cows.

4 **A shed has bicycles (2 wheels) and tricycles (3 wheels). There are 13 of them with 33 wheels.**

(a) The number of tricycles is ☐

(b) The number of bicycles is ☐

Working: _____

Start here

Keep going. Check your work.

5 Mei had some stickers. She gave half of them to Ali. Then she bought 12 more. Then she lost 7. Now she has 25 stickers. How many did she have at first?



6 Kumar thinks of a number. He doubles it, adds 14 and then divides by 4. His answer is 10. What is his number?

Working: _____

7 Ali, Ben, Mei and Siti each have a different pet: a cat, a dog, a fish or a rabbit. Use the grid. Write ✓ for yes and × for no.

	cat	dog	fish	rabbit
Ali				
Ben				
Mei				
Siti				

(a) Clues: Ali's pet barks. Ben's pet has no fur. Mei's pet does not hop. Pets: Ali ____, Ben ____, Mei ____, Siti ____

Start here

Nearly there. You can do it.

8 **Aisha is 9th from the front and 5th from the back of a queue.**

(a) How many people are in the queue? ☐

(b) In another queue there are 25 people. Kumar is 8th from the front. He is ☐ th from the back.

Draw the queue here.

9 **Siti and her mother are 44 years old altogether. In 6 years her mother will be 3 times as old as Siti. How old is Siti now?** ☐

Draw a bar model here.

10 **Ravi has 9 coins. They are 20¢ coins and 50¢ coins. They are worth $3.00 altogether. How many 50¢ coins does he have?** ☐

Working: _____

Think harder

Hints are upside down. Try first!

1 **A hawker centre has 14 tables. Some seat 4 people and some seat 6 people. Together they seat 68 people.**

(a) The number of tables that seat 6 is ☐

(b) The fewest and the most people that the 14 tables could seat: ☐ and ☐

Working: _____

_____

Hint. Suppose all 14 tables seat 4 people. How many seats are missing?

2 **Five children sit in a row of 5 seats, numbered 1 to 5 from the left. Aisha is in seat 3. Kumar is at one end, not next to Ravi. Mei is to the right of Ravi. Ben is not next to Aisha.**

(a) Seats 1 to 5, in order: ☐ ☐ ☐ ☐ ☐

Try: _____

_____

Hint. Where can Kumar sit? Try each end and check every clue.

3 **A grandfather is 7 times as old as his grandson. In 12 years, he will be 3 times as old. How old is the grandson now?** ☐

Draw a bar model here.

_____

_____

Hint. The age gap stays the same. Make a bar model for now and for 12 years later.

Go further

A weekend page. Big challenges. Hints are upside down.

1 **How many whole numbers from 1 to 200 have digits that add up to 5?** ☐

List: _____

Hint. List them in order: 1-digit, 2-digit, then the 100s.

2 **Ali has 3 times as many 20¢ coins as 50¢ coins. His coins are worth $4.40 altogether. How many coins does he have in all?** ☐

Working: _____

Hint. Make groups: 3 twenty-cent coins and 1 fifty-cent coin. What is a group worth?

3 **In a line of children, Ben is 11th from the left. Mei is 8th from the right. There are 5 children between Ben and Mei. How many children could be in the line? Find both answers.**

☐ **or** ☐

Draw the line here.

Hint. Mei can be on the right of Ben or on the left of Ben.

4 **Beyond Level 4: Number cycles**

The ones digits of 3, 3 × 3, 3 × 3 × 3 and so on go round in a cycle.

(a) The ones digit of 3 × 3 × 3 × 3 × 3 (five 3s) is ☐

(b) The ones digit of 20 threes multiplied together is ☐

Ones digits: _____

Hint. Write the ones digits of the first 6 products. Look for the cycle.

✔ Check yourself

No hints. Try each one on your own.

1 **A garage has cars (4 wheels) and bikes (2 wheels). There are 10 of them with 28 wheels. How many cars are there?** ☐

Working: _____

2 **Ali thinks of a number. He adds 8, doubles it and takes away 6. His answer is 30. What is his number?** ☐

Working: _____

3 **Siti is 6th from the front and 9th from the back of a queue. How many people are in the queue?** ☐

Working: _____

4 **Two next-door numbers multiply to make 272. What are they?** ☐ **and** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Factor hunters	☐	☐	☐	☐	/ 4	
2	Big number puzzles	☐	☐	☐	☐	/ 4	
3	Operation detectives	☐	☐	☐	☐	/ 4	
4	Fraction frontier	☐	☐	☐	☐	/ 4	
5	Decimal detectives	☐	☐	☐	☐	/ 4	
6	Angle turns	☐	☐	☐	☐	/ 4	
7	Area architects II	☐	☐	☐	☐	/ 4	
8	Symmetry and nets	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Puzzle Day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 4.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 10 · Puzzle Day

Your turn
page 6

Answer: 3
Why: Suppose all 10 are bees: 60 legs. That is 6 short. Each swap to a spider adds 2 legs, so 3 swaps. There are 3 spiders and 7 bees. Check: 3 × 8 + 7 × 6 = 66.

START HERE

1
page 7

Answer: 22 and 23
Why: 20 × 21 = 420 is too small. 25 × 26 = 650 is too big. 22 × 23 = 506. Use each try to decide the next one: 21 × 22 = 462 is too small, so go up.

2
page 7

Answer: (a) 3 (b) 2 pens and 7 notebooks
Why: Try the number of notebooks: 1 notebook leaves $36, which is 12 pens. 2 leaves $31 (no). 3 leaves $26 (no). 4 leaves $21, which is 7 pens. 5 leaves $16 (no). 6 leaves $11 (no). 7 leaves $6, which is 2 pens. 8 notebooks cost $40 and leave $1 (no). So the ways are 12 pens and 1 notebook, 7 pens and 4 notebooks, and 2 pens and 7 notebooks.

3
page 7

Answer: (a) 34 (b) 16 (c) 8
Why: (a) 17 × 2 = 34. (b) 50 − 34 = 16. (c) 16 ÷ 2 = 8 swaps, so 8 cows and 9 hens. Check: 8 × 4 + 9 × 2 = 32 + 18 = 50.

4
page 7

Answer: (a) 7 (b) 6
Why: Suppose all 13 are bicycles: 26 wheels. That is 7 short. Each swap to a tricycle adds 1 wheel, so 7 tricycles and 13 − 7 = 6 bicycles. Check: 7 × 3 + 6 × 2 = 33.

5
page 8

Answer: 40
Why: Work backwards. Undo the loss of 7: 25 + 7 = 32. Undo the 12 she bought: 32 − 12 = 20. Undo giving away half: 20 × 2 = 40. Check: half of 40 is 20, 20 + 12 = 32, 32 − 7 = 25.

6
page 8

Answer: 13
Why: Work backwards. 10 × 4 = 40. 40 − 14 = 26. 26 ÷ 2 = 13. Check: 13 × 2 = 26, 26 + 14 = 40, 40 ÷ 4 = 10.

7
page 8

Answer: (a) Ali dog, Ben fish, Mei cat, Siti rabbit
Why: Ali's pet barks, so it is the dog. The fish is the only pet with no fur, so Ben has the fish. A rabbit hops, so Mei does not have the rabbit. The cat and the rabbit are left. Mei has the cat and Siti has the rabbit.

8
page 9

Answer: (a) 13 (b) 18
Why: (a) There are 8 people in front of Aisha and 4 behind her, and Aisha: 8 + 1 + 4 = 13. (b) There are 7 people in front of Kumar, so 25 − 8 = 17 behind him. He is 17 + 1 = 18th from the back.

9
page 9

Answer: 8
Why: In 6 years, each is 6 years older, so the total is 44 + 6 + 6 = 56. Siti is 1 unit and her mother is 3 units: 4 units = 56, so 1 unit = 14. In 6 years Siti is 14, so she is 8 now. Her mother is 36 now. Check: 36 + 6 = 42 = 3 × 14.

10
page 9

Answer: 4
Why: Suppose all 9 are 20¢: 180¢. That is 120¢ short. Each swap to a 50¢ coin adds 30¢. 120 ÷ 30 = 4 swaps. So 4 coins of 50¢ and 5 coins of 20¢. Check: 200¢ + 100¢ = 300¢.

THINK HARDER

1
page 10

Answer: (a) 6 (b) 56 and 84
Why: (a) If all 14 tables seated 4, that would be 56 seats. 68 – 56 = 12 short. Each table that seats 6 adds 2, so 6 tables seat 6 and 8 tables seat 4. Check: 36 + 32 = 68. (b) The fewest is 14 × 4 = 56 and the most is 14 × 6 = 84.

2
page 10

Answer: (a) Ben, Ravi, Aisha, Mei, Kumar
Why: Kumar is in seat 1 or seat 5. Try seat 1: then Ravi is in seat 3 or later and Mei is to his right, so Ravi is in seat 4 and Mei in seat 5, but then Ravi must not be next to Kumar (fine) and Ben has to take seat 2, next to Aisha. That is not allowed. So Kumar is in seat 5. Ravi is not in seat 4, so Ravi is in seat 2 (seat 3 is Aisha). Mei is in seat 4, and Ben is in seat 1, not next to Aisha. All clues fit.

3
page 10

Answer: 6
Why: The gap in their ages never changes. Now it is 6 units (7 – 1). In 12 years it is 2 units (3 – 1). So 6 units = 2 units of the later bars. If the grandson is g now, the grandfather is 7 × g, and 7g + 12 = 3 × (g + 12), so 4g = 24 and g = 6. Check: 42 and 6 now, 54 and 18 in 12 years, and 54 = 3 × 18.

GO FURTHER

1
page 11

Answer: 11
Why: 1-digit: 5. 2-digit: 14, 23, 32, 41, 50. 3-digit, starting with 1: the other 2 digits add to 4: 104, 113, 122, 131, 140. 200 has a digit sum of 2. In all 1 + 5 + 5 = 11.

2
page 11

Answer: 16
Why: A group of 3 coins of 20¢ and 1 coin of 50¢ is worth 60¢ + 50¢ = 110¢. $4.40 = 440¢ and 440 ÷ 110 = 4 groups. Each group has 4 coins, so he has 16 coins: 12 coins of 20¢ and 4 coins of 50¢.

3
page 11

Answer: 12 or 24
Why: If Mei is to the right of Ben, she is 6 places after him: 11 + 6 = 17th from the left. She is 8th from the right, so there are 17 + 7 = 24 children. If Mei is to the left of Ben, she is 11 – 6 = 5th from the left, so there are 5 + 7 = 12 children. Both fit: the 5 children in between are places 6 to 10 in the line of 12, or places 12 to 16 in the line of 24.

4
page 11

Answer: (a) 3 (b) 1
Why: The products are 3, 9, 27, 81, 243, 729, 2187. The ones digits are 3, 9, 7, 1, 3, 9, 7 and go round in a cycle of 4. (a) The 5th product ends in 3. (b) 20 is 5 whole cycles of 4, so the 20th product ends in the last digit of the cycle, which is 1.

CHECK YOURSELF

1
page 12

Answer: 4
Why: Suppose all 10 are bikes: 20 wheels, 8 short. Each swap to a car adds 2 wheels, so 4 cars and 6 bikes. Check: 16 + 12 = 28.

2
page 12

Answer: 10
Why: Work backwards. 30 + 6 = 36. 36 ÷ 2 = 18. 18 − 8 = 10. Check: 10 + 8 = 18, 18 × 2 = 36, 36 − 6 = 30.

3
page 12

Answer: 14
Why: There are 5 people in front of Siti and 8 behind her. 5 + 1 + 8 = 14.

4
page 12

Answer: 16 and 17
Why: 15 × 16 = 240 is too small and 17 × 18 = 306 is too big. 16 × 17 = 272.